Computer Vision

Code :- 1TA0503

Co2 :- Analytical Problem Solving.

Name :- A. Lakshmi Devi

Reg no :- ~~1925~~ 192525176.

Thresholding and Intensity-level slicing - PCB Inspection

Given:

Original pixel values:

$r = [25, 60, 95, 130, 165, 200, 235]$

a) Binary threshold transformation

If $r \geq 128$ Output $s = 255$ Otherwise, $s = 0$

Inputs	Outputs
25	0
60	0
95	0
130	0
165	255
200	255
235	255
	255

$[0, 0, 0, 255, 255, 255, 255]$

b) Intensity-level slicing.

Pixels Intensity range 90-170 become 255

Inputs	Outputs
25	25
60	60
95	255
130	255

200
235

200
235

[95, 60, 255, 255, 255, 200, 235]

c) Number and Percentage of highlighted pixels.

Three Pixels (95, 130 and 165)

Percentage = $\frac{3}{7} \times 100$

$\boxed{42.86\%}$

d) Mean intensity.

Original row:

$$\text{Mean} = \frac{95+60+95+130+165+200+235}{7}$$

$$= \frac{910}{7} = \boxed{130}$$

Sliced Output :-

$$\text{mean} = \frac{95+60+255+255+255+200+235}{7}$$

$$= \frac{1085}{7} = \boxed{183.57}$$

e) Which transformation is more appropriate
Intensity-level Slicing is more appropriate
because it highlights Only the pixels
in the required intensity band of 90-170.

Piecewise - Linear Contrast stretching
- Ray Image Enhancement.

Given:

- $r_1 = 70$, $S_1 = 20$
- $r_2 = 140$, $S_2 = 230$
- $L = 256$ So the Intensity range is 0-255

a) Equations and Slopes of the three Segment.

The general Slope formula is:

$$m = \frac{S_2 - S_1}{r_2 - r_1}$$

Segment 1: 0-70

Using Points (0, 0) and (70, 20)

$$m_1 = \frac{20-0}{70-0} = \frac{2}{7} = 0.2857.$$

$$S = 0.2857r, \quad 0 \le r \le 70$$

Segment 2: 70-140

$$m_2 = \frac{230-20}{140-70} = \frac{210}{70} = 3$$

$$S = 3r - 190, \quad 70 \le r \le 140$$

Segment 3: 140-255

Using Points (140, 230) and (255, 255)

$$m_3 = \frac{255-230}{255-140} = \frac{25}{115} = 0.2174$$

b) Compression and expansion.

Segment	Slop	Effect
0-70	0.2857	Compression.
70-140	3	Expansion
140-255	0.2174	Compression.

A slope greater than 1 expands contrast; a slope less than 1 compresses contrast.

c) Does the transformation improve visibility?

Yes. It improves the visibility of diagnostic details in the 70-140 range because the contrast is expanded by a factor of 3.

3) Spatial filtering - Impulse Noise removal.

Given 3x3 neighbourhood:

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 54 \end{bmatrix}$$

The Centre Pixel 255 is Salt noise.

a) 3x3 Mean (box) filter.

formula:

$$\text{Mean} = \frac{\text{Sum of all 9 Pixels}}{9}$$

$$= \frac{52+55+58+54+255+57+53+56+54}{9}$$

$$= \frac{699}{9} = \boxed{77.67}$$

b) Weighted - average filter.

Given mask:

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Multiplying each Pixel by its Corresponding weight:

$$= \frac{52+110+58+108+1080+114+53+110}{16}$$

$$= \frac{1686}{16} = \boxed{105.375}$$

c) Median filter.

Arrange all nine values in ascending order.

52, 53, 54, 55, 56, 57, 58, 59, 255

The middle (fifth) value is 56

Median Output = 56

d) Absolute error of each filter.

The noise-free reference is the average of the eight neighbours.

$$R = \frac{52+55+58+54+57+53+56+59}{8}$$

$$R = \frac{444}{8} = 55.5$$

Absolute error formula;

$$E = |\text{filter Output} - R|$$

Filter	Output	Absolute error
Mean	77.67	22.17
weighted average	105.375	49.875
Median	56	0.5

c) Most appropriate filter:
The Median filter is the most appropriate for removing Salt-and-pepper noise while preserving edges.

4) Edge detection - Sobel and Prewitt Gradient Analysis:
Given image region:

$$A = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$$

a) Sobel operator: Calculate G_x and G_y.
Given: $G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

for G_x:

$$G_x = (-10+10) + (-20+180) + (-10+90)$$

$$\boxed{G_x = 240}$$

for G_y:

$$G_y = (-10-20-10) + (10+180+90)$$

$$\boxed{G_y = 240}$$

b) Gradiant Magnitude and approximation.

$$M = \sqrt{G_x^2 + G_y^2}$$

$$M = \sqrt{240^2 + 240^2} = \sqrt{115200}$$

$$\boxed{M = 339.41}$$

$$M \approx |G_x| + |G_y|$$

$$E = \frac{|480 - 339.41|}{339.41} \times 100$$

$$\theta = 41.42\%$$

c) Gradient direction and edge Orientation.

$$\alpha = \tan^{-1}\left(\frac{Gy}{Gx}\right)$$

$$\alpha = \tan^{-1}\left(\frac{240}{240}\right)$$

$$\boxed{\alpha = 45^\circ}$$

5) Global thresholding - Iterative and Otsu Segmentation.

Intensity T_k	0	1	2	3	4	5	6	7
frequency M_k	3	6	4	1	1	5	8	4

Total Pixels:

$$N = 3+6+4+1+1+5+8+4 = 32$$

a) Global Mean Intensity.

$$M_G = \frac{\Sigma r_k M_k}{N}$$

$$M_G = \frac{0(3)+1(6)+2(4)+3(1)+4(1)+5(5)+}{32}$$

$\frac{105}{32}$

$m_G = 3.8105$

b) Basic Iterative global thresholding

Initial threshold:

$T_0 = m_G = 3.8125$

Divide

C_1 : Intensity values less than or equal to T!

C_2: Intensity values greater than T.

I_1:-

$T_0 = 3.8125$ level = 0, 1, 2, - - -

$$m_1 = \frac{0(3) + 1(6) + 2(4) + 3(1)}{3+6+4+1}$$

$$m_1 = \frac{17}{14} \approx 1.2143$$

C_2 level 4, 5, 6 and 7

$$m_2 = \frac{4(1) + 5(5) + 6(8) + 7(4)}{1+5+8+4}$$

$m_2 = 105/18 = 5.8333$

$$T_1 = \frac{m_1 + m_2}{2}$$

$$T_1 = \frac{1.2143 + 5.833}{2} = \boxed{3.5238}$$

Threshold

$$\Delta T = |3.5238 - 3.8195| = 0.0887.$$

c) Otsu's Method.

$$\sigma_B^2(k) = P_1(k) P_2(k) [m_1(k) - m_2(k)]^2$$

for $k = 3$

$P_1 = 13/32 = 0.40625$ $P_2 = \frac{19}{32} = 0.59375$

$m_1 = 14/13 = 1.0769$, $m_2 = \frac{108}{19} = 5.6842$

$$\boxed{\sigma^2_B(2) = 5.1202}$$

d) Optimum otsu threshold and binary mapping

Threshold k	Blw class variance.
2	5.1202
3	5.2506
4	5.1354

Maxi vari is $k = 3$

e) Compare the two methods and assess the histogram.

. Iterative thresholding: $T = 3.5238$

. Otsu thresholding: $k^* = 3$